

volved in their complete disassembly and subsequent reassembly. Therefore the other option for clearing out these liquids is to accelerate evaporation, which can be done by both drying out the air around the device as much as possible and applying heat and then waiting.

Tips to save an electronic device from a water death

The first and most important thing is to turn off your gadget and, if possible, remove the battery and any other parts that can be taken out (SIM card, memory card, headphones and the like), along with the phone case and cover. If the power is off, there will be no current to short the circuit. This will improve the chances of recovering the device.

Remove all water. After the power is switched off, try to remove as much water as possible from the device, either by gently shaking it out, wiping it or removing a few panels or parts to get these really dry. Grab a soft cloth (towel or t-shirt) and soak up all excess moisture. Use a cotton swab to get around the buttons and headphone jack.

Use Q-tips and alcohol. If water is stuck in certain small parts of the device, dab a Q-tip into some alcohol (or vodka) and apply it to those parts of the device. This will work well for the internal parts of many gadgets that have circuit boards. Alcohol will help speed up evaporation, which happens when the device sits out in open air.

Use desiccants. Unfortunately, many a time people use methods for drying wet electronics that are either ineffective or may damage the device. For instance, dried grains such as rice can absorb liquids, so some people use this as desiccant to dry out a device. However, rice is relatively ineffective at absorbing liquid from the air and may not be of much help.

Additionally, since heat helps water evaporate, people have at times baked their devices in ovens or with hair dryers, resulting in melted

components and burned circuits (memory circuits such as ROM and RAM are fairly susceptible to damage from over-heating).

Other absorbers like silica gel (kitty litter) and instant couscous are known to work in some cases, besides the plain old air and sunshine. However, a new scientific study on an emergency electronic device rescue service released recently shatters the widespread myth that placing a wet smartphone in a container of rice is an effective way to dry it.

The optimal window of opportunity to restore a wet electronic device is less than 48 hours, and it is always best to act as soon as possible.



Thirsty Bag with iPhone (Image courtesy: eustore.litix.com)

sible. The study not only proves that rice is an ineffective drying agent, but that it is actually less effective than simply leaving the device out in the open to dry naturally on its own—but neither method will thoroughly dry the device.

Thirsty Bag. To help in drying out small devices like mobile phones and wrist bands, engineers have come up with Thirsty Bag, which is a sealed container that contains molecular sieve desiccant. Place a drenched phone, camera or any other device in it; after a while the liquid will get absorbed into the desiccant and leave the device dry. To help the process, heat the package a bit using an electric warming pad or similar device. But do not use an oven, as it is very easy to over-heat and damage an otherwise restorable device.

Thirsty Bag is useful, but is meant for smaller devices. For drying out laptops and other items that will not fit in the bag, you will need

canned or compressed air, heating pad or other heat source and sealed bag or airtight plastic container that is just large enough to hold your device.

With these tools handy, remove the battery, hard drive and optionally RAM. Then, unscrew the bottom cover and remove it so that the guts of the system are exposed. Use a few wads of toilet paper to wedge the screen open. Blow canned air in and around the device to rid it of as much liquid as possible.

After this is done, prepare the plastic container by placing the desiccant in the bottom and covering it with some paper towels. Place the laptop and its removed components in the container, propping these up so air can flow around as much as possible. Seal the container to prevent humid air from entering it.

The last step is to place the container in a warm place, or on a heating pad, and let it stay there for at least a few days. Ensure that the temperature does not exceed the recommended storage temperature for your device (check your device's specifications). Depending on the degree of fluid penetration, you might need to keep the device in there for up to a week to ensure it is fully dry.

If the device is still not working and is not covered by any warranty, you can rinse the device with pure distilled water and dry it again. When liquids like soda or juice dry, these leave residual salts behind that can continue to short out a circuit.

Feel free to liberally dunk the device in copious amounts of pure distilled water (ensure it is distilled and not mineral water) so it flows in and around it, followed by draining the device, opening it up and placing it in the desiccant chamber again to dry. Doing this should clear out residual salt contaminants left behind from the initial liquid, and potentially clear any shorts these may be causing. But do this as a last resort if an initial drying does not work, and if the device is not covered by any warranties. **EFY**